

2. Inoculation of mycelium on agar substrate

To make pure mycelium films on a petri-dish, we need another experimental procedure than the growth of mycelium blocks.

Step by Step

Materials

- Mother culture
- Agar substrate
 - For 1 liter of distilled water :
 - 20g of agar
 - 20g of malt extract
 - 2g of yeast extract
 - 1g of peptone
- Scale
- Petri-dish
- Flask
- Autoclave tape, parafilm
- Magnetic stirrer
- Scalpel
- Lighter
- Ethanol
- Parafilm
- Pipette

Procedure

1. Prepare the substrate by weighing the ingredients on a scale. Take 4 different spoons for 4 different ingredients. The little round dot on the right of the scale is used as a indicator of how balanced the scale is (i.e. well-balanced if dot is in the middle). If not balanced, adjust the scale with the red screws in the bottom.



2. Mix the distilled water and the substrates in the flask. First put a little bit of water to facilitate dissolving.
3. Do not completely close the lid of the flask and put autoclave tape (turns black when sterilization is done).



4. Put the flask in the autoclave machine.
5. When the mix is ready, stir it once with a magnetic stirrer (~ 56 for turn, ~70degrees for temperature)



6. Turn on the light and the ventilation of the hood. UV light can be turned on for a few minutes for disinfection (should avoid contact with eyes and skin).



7. Sanitize the hood with ethanol.
8. Sanitize the scalpel with a lighter.
9. Prepare the petri-dishes. Try to stay away from the edge to prevent contamination.
10. Before the mix cools down, pipette 25ml of mix for 1 petri-dish.

11. Wait until the mix jellifies (~2-3 minutes). Check by gently shaking the dish right and left.
12. Write your *name – mycelium specie – date of preparation* on the petri-dishes (not the lid) that are waited to be inoculated.
13. Get the mother culture. Check that the scalpel has cooled down and cut pieces of the culture (including the jelly) in a consistent size of about **4x4xmm cubes**.



14. Place the cubes in the middle of the petri-dishes, close the lid and seal it with parafilm.
15. Put the rest of the petri-dishes that only contain agar jelly substrate back in the plastic bag. The petri-dishes should be upside down to prevent condensation on the lid.
16. Write your *name – content of the substrate (e.g. AMYP for this one) – date of preparation* on the plastic bag and put it in the refrigerator. The substrate can be used until 1 month after preparation.
17. Put the inoculated mycelium in the growth chamber. Film should be ready in a week.
18. Wash and sanitize the equipment used with ethanol.